Divyansh Maurya

New York City | +1 6465544479 | dm6602@nyu.edu | LinkedIn | GitHub | Personal Website

SUMMARY

AI Engineer with **2+ years** of software engineering experience shipping **LLM-powered applications and agents**: **tool use / function calling**, **RAG**, **agent orchestration (LangGraph)**, **guardrails**, and **evaluation-driven iteration**, backed by **deep learning in PyTorch**. Owns **Python/TypeScript** AI services **end to end**, from data pipeline to deployed product.

EDUCATION

New York University **September 2025 - May 2027**
Master of Science, Computer Science *New York City, USA*
Coursework: Generative AI, Deep Learning, Machine Learning, Artificial Intelligence, Big Data, Software Engineering

National Institute of Technology, Tiruchirappalli **August 2019 - May 2022**
Master of Computer Applications *India*

Banaras Hindu University **August 2015 - May 2018**
Bachelor of Science (Hons.), Computer Science *India*

TECHNICAL SKILLS

LLMs & Agents: Claude & Gemini APIs, Prompting, Function Calling, RAG, LangGraph, Guardrails, LLM Evals
ML & Deep Learning: PyTorch, CNN Ensembles (EEGNet, ShallowConvNet), LSTMs, Few-Shot Calibration, Forecasting
Languages & Serving: Python, TypeScript, JavaScript (ES6+), SQL, FastAPI, Node.js, REST / OData, React
Data & Retrieval: pgvector, ChromaDB, Kafka, Spark, Delta Lake, AWS (S3, RDS, DynamoDB, Glue, Athena)
Cloud & DevOps: GCP, SAP BTP, Docker, Jenkins, CI/CD, Git, Linux, AI Coding Agents, Unit & Integration Testing

EXPERIENCE

MSITEK
AI Engineer Intern *Princeton, New Jersey* · **June 2026 - August 2026**
- Owned **SMart Retail AI** end to end, from design to deployment: a production **AI platform** on SAP BTP cloud that unifies **8 AI and predictive engines** (forecasting, anomaly detection, replenishment, pricing, personalization) in one pipeline.
- Integrated **Anthropic Claude** as the **LLM reasoning layer** for **agentic alert triage**, with **guardrails** grounding outputs in platform data; automated **76% of checkout-integrity alerts (367/482)**, surfacing **EUR 7.3K+** in value.
- Ran **evaluation-driven model selection**: trained a **ridge-regression predictive model** and **benchmarked** it against the baseline forecaster on **120K+ records** before promoting it.
- Built the serving layer: **CDS models**, **OData V4/REST APIs**, a **Node.js** POS/ERP ingestion service, and a **React** PWA.
- Used **AI coding agents** to write code and tests to spec (**68/68** unit and integration tests passing); owned documentation and knowledge transfer, and shipped a product **approved by SAP for the SAP Store**.

Fastenal
Software Development Engineer *Bangalore, India* · **July 2022 - March 2024**
- Designed and deployed **RESTful APIs** and third-party integrations serving high-volume production traffic at **99.9% uptime**, boosting customer engagement **30%**; monitored latency with Dynatrace and Kibana and led **root-cause analysis**.
- Built and maintained CI/CD pipelines with Jenkins, Docker, and TeamCity, cutting deployment failures **35%**.

Software Development Engineer Intern *Bangalore, India* · **January 2022 - May 2022**
- Optimized asynchronous **Node.js** business logic in a distributed microservices supply chain app, cutting latency **40%**.

PROJECTS

Brain-Controlled Wheelchair: Agentic Real-Time AI on EEG Brainwave Patterns **February 2026 - April 2026**
- Built a **LangGraph agent loop** that runs **signal-quality self-checks** before acting and uses **pgvector retrieval** to ground each prediction's explanation, in a **real-time, event-driven pipeline** (FastAPI, Kafka, Spark, Delta Lake).
- Trained an **EEGNet + ShallowConvNet** ensemble in **PyTorch**, pre-trained on **153 subjects across 3 EEG datasets** with 50-shot calibration, reaching **88.1%** accuracy on BCI IV-2a; documented **benchmarks** in an IEEE-format paper.

SubNest: LLM Lead-Conversion Agent for Real Estate Brokerages **subnest.ai**
- Built an **LLM concierge agent** on the **Gemini API** that moves leads through a **7-stage conversational pipeline**: **intent extraction**, property matching, and broker handoff, driven by prompt-engineered instructions and session-aware context.
- Added a **real-time voice mode** on the **Gemini Live API** with **tool calling** that captures lead data from speech and advances pipeline stages automatically; shipped as a React + TypeScript app on Vercel.

ACHIEVEMENTS

- Represented NYU at **NASA Space Apps Challenge 2025**.
- Invited **guest speaker** at Softrec NITT AGM '26, NIT Trichy; mentor to NIT Trichy and BHU students.
- Secured **All India Rank 76** in NIMCET 2019 (NIT MCA Common Entrance Test).